SC4001 · PROJECT E

Catching AI hallucinations in picture captioning

Goal: Build a small detector that spots hallucinations from AI's hidden layer activations.

EXAMPLE

"A man throwing a frisbee in a park, next to a ~~dog~~."

green in the photo ~~red~~ not in the photo: a hallucination

Proposed process

1

Get photos

Use COCO dataset - Large set of photos with labelled captions.

`~4,000 photos` `70 / 15 / 15 split`

out → photos + human object lists

2

Let the AI caption each picture

While the AI writes, we save its hidden layer activations.

`Qwen2-VL-2B · frozen, not trained`

out → captions + internal numbers for each word

3

Find Mistakes

For each picture, we compare the list of object in AI caption vs human one. Not on the list means a mistake.

`CHAIR matching` `hand-check ~100`

out → each object word labelled real or made up

4

Train the detector

Using the mistaken objects pictures, and the hidden layer values for those, we train a small detector to perceive pattern in values.

out → detector that says real or made up

5

Plug the detector in to fix descriptions

We try two ways.

AFTER WRITING

Feedback loop

1. The AI finishes the description.
2. The detector flags "dog".
3. We tell the AI: "dog is likely not there".
4. The AI rewrites it.

WHILE WRITING

Early warning

1. The AI is about to write "dog".
2. The detector checks first.
3. It says "made up".
4. We block "dog". The AI picks another word.

6

Test on new photos

We use photos from our test set the detector never saw.

`AUROC · F1` `CHAIR before vs after` `real objects kept`

7

Compare

Is our detector better than simple checks? Does fixing help more than random flags? How close is it to perfect flags?

`word confidence` `object detector` `random flags` `perfect flags`

Runs on a free Google Colab T4 GPU. Data: COCO val2014. Model: Qwen2-VL-2B-Instruct.